

- 3 Modern gravity meters can measure g , the acceleration of free fall, to a high degree of accuracy. The principle on which they work is of measuring t the time of fall of an object through a known distance h in a vacuum.

- (a) Assuming that the object starts from rest, deduce an expression for g in terms of t and h .

$$s = ut + \frac{1}{2}at^2$$

$$h = 0 + \frac{1}{2}gt^2$$

$$g = \frac{2h}{t^2}$$

[1]

- (b) State Newton's law of gravitation relating the force F between two point objects of masses m and M , their separation r and the gravitational constant G .

$$F = \frac{GMm}{r^2}$$

Newton's Law of Gravitation states that one point object will attract the other point object with a gravitational force that is directly proportional to the product of the masses of the particles and inversely proportional to the square of the distance between them, where G is the constant of proportionality..

$$= \frac{(6.67 \times 10^{-11}) \times (1) \times (2800 \times \frac{4}{3} \pi (2000)^3)}{(4000)^2}$$

$$= 3.91 \times 10^{-4} \text{ N}$$

[1]

- (c) Fig. 3.1 shows a standard kilogram mass at the surface of the Earth and a spherical region S of radius 2000 m with its centre 4000 m from the surface of the Earth.

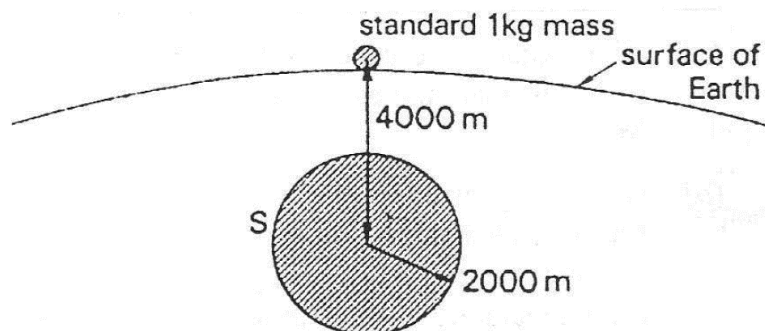


Fig. 3.1 (Not to scale)

Region S is made up of rocks with density 2800 kg m^{-3} . Calculate the force the matter in region S exerts on the standard mass.

$$\begin{aligned}
 F &= \frac{GMm}{r^2} \\
 &= \frac{(6.67 \times 10^{-11}) \left(\frac{4}{3} \pi (2000)^3 (2800) \right) (1)}{4000^2} \\
 &= 3.91 \times 10^{-4} \text{ N}
 \end{aligned}$$

force = N [2]

- (d) Calculate the difference in the force on the standard mass, if region S consists of oil with density 900 kg m^{-3} instead of rock.

The difference between the values of the forces in (c) and (d) is due to the density of S.

$$\begin{aligned}
 F_{oil} &= \frac{900}{2800} 3.91 \times 10^{-4} \text{ N} \\
 &= 1.26 \times 10^{-4} \text{ N}
 \end{aligned}$$

Or any other correct means of calculation

$$\begin{aligned}
 \Delta F &= 3.91 \times 10^{-4} - 1.26 \times 10^{-4} \\
 &= 2.65 \times 10^{-4} \text{ N}
 \end{aligned}$$

difference in force = N [2]

- (e) In oil prospecting, the gravity meter is used to predict where deposits of oil are under the surface of Earth. Use your answers in (d) to suggest one necessary condition in using the gravity meter.

The gravity meter has to be precise to 5 significant figures to detect the difference

in forces.

..... [1]

[Total: 7]